

Company: IDFC

Job Profile: Application Engineer

Offer Type: I+P

Internship Stipend:

CTC: 18.10 (Fixed 14LPA + Joining Bonus 2L + Variable 2.10)

Location: Mumbai/Bangalore/Hyderabad

Process Date: OA - 28/07 & Interviews - 30/07

Placement Process:

The company first conducted a pre-placement talk, which was followed by an online assessment few days later.

After that 4 interview rounds were conducted on 30th July.

1. Online Assessment [135 students appeared for it]

This was conducted on the HirePro platform & duration was for 75mins. The test consisted of 3 sections, each with a time limit. The first section had 15 questions and was aptitude based wherein general questions on logical reasoning were asked. The second section had 25 technical questions based on Excel, CCN, OS, OOPs and Basic Statistics. The third section had 2 coding questions which were quite easy.

- Q1) How many elements are correctly sorted.

Correct place means Rightly indexed when seen from a Sorted perspective.

Ex: 1 3 2 4 5

sorted is 1 2 3 4 5

1 and 4 are in the correct place , whereas 3 and 2 are not in the correct place.

So the result would be

2, 3

Since 2 are incorrect and 3 are correct

- Q2) Had to find the number of golden number in a given range where the golden number meant the sum of the digits of a number - max number = max number

Ex: 325 is a golden number

$(3+2+5) - 5 = 5$

2. Technical Round 1 [41 students appeared]

Initially 47 to 50 students were there but then many got selected in barclays so finally 41 students had appeared for this round.

This round lasted for me around 45 mins to 1hr. I was the 1st person that was to be interviewed in my panel. It started with my introduction and then he directly jumped into a DSA problem it was related to cache so we had a buffer size of N and whenever we pushed an element it had a certain TTL (time_to_live) so

suppose an element entered the buffer at 10:20 and its TTL was 2mins then at 10:23 it was supposed to be popped out of the cache and make room for other elements to enter.

Initially i started with a hashmap solution but as soon as i started explaining i got the Priority Queue solution using heap here since my primary language was python he was checking my syntax or the function that i was mentioning with GFG and since in GFG the priority queue implementation was not the one using heap he started asking what is heap, heapify and balanced binary tree which i was able to explain it all.

Then he started grilling me on GenAI and bit of ML since my entire resume was based on that

Q1) Difference between GenAI and unsupervised learning (**Generates new content and solely revolves around the input to find structure**)

Q2) What is the output of unsupervised learning models (**Clusters, Dimensionality Reduction**)

Q3) Architecture of GenAI (**Input layer + Tokenization + Embedding + Transformer + Attention Mechanism+ Decoder/output generator**)

Here i gave a brief overview of each layer and was trying to keep it as detailed as possible since i didnt want him to ask any more question in depth over here since i hadnt done transformers architecture in detail.

Q4) How does it matches the user context and generates an appropriate content (**Semantic Similarity**)

Q4) What is RAG

Q5) In a glimpse explain what is prompt engineering

Q6) Since i had explained RAG with the example of a medical chatbot he asked if a user prompts and says he cant concentrate in his work but the bot doesn't have any idea, as concentrate is not there in its DB then what will i do to give a proper response, initially i had said if at any point the bot cannot give a proper advice then i would have a default response wherein it will provide a general physician and he can take his concern forward with him but this wasn't what he had expected he said that the chatbot should ask few more questions on what he is feeling so that he can reach a conclusion and then provide a solution

He did ask few more questions based on AIML which i am not able to recollect but then it ended with me asking a question based on AIs usage in IDFC and one common question

3. **Technical Round 2** [22 students]

This round lasted for around 40 mins for me

Here i didnt even feel i would get selected as the people who had went previous to me were all coming out demotivated since they were asked questions based on system design, AWS, CCN and i hadnt prepared in that terms at all but i was lucky since i was not asked such Qs

DSA : [Valid Parenthesis](#)

Resume Based:

1. Asked me to explain one of my data analytics project it was dashboard based then asked few questions based on that
2. Since I had SQL in my resume they asked me what kind of projects I have done. I explained one, then they asked few simple queries based on the project explained, they were based on aggregate, group by and one subquery related cant remember the exact questions but all of them was very easy.
3. What is Jupyter Notebook and why use it when you have PyCharm or other IDEs
4. Then they asked which OS did i use since OS was mentioned in my resume (**I had said windows on my personal laptop and it was ubuntu on college PC**) so they started asking some linux commands and more related to that

5. Since my primary language was python they asked few questions based on that like types of exception handling in python and revolved around that more
6. Then they gave me a usecase kinda question like suppose there would be finger scanner for storing many peoples data what all would be your approaches. This was kind of an open ended question they wanted to see my how i would approach it **(Here i would say ask whatever question you can possibly ask, coz more than the answer they want to see what you think so i asked what kind of data will be stored they said 0s and 1s i instantly started thinking in terms of FOSIP though i couldnt think much but mentioned few of the things that i could remember like grayscaling image optimization, thresholding and RLE and as i kept answering i even asked do we need to even process on the storage part for that they said you may so i mentioned a bit about the server and load balancing since they had said millions of data will be there and so on to retrieve as and when needed so but this a gist there was a detailed conversation on this)**

There were few more questions but were based on fundamentals and were answerable and in the ending asked the same question as before.

4. **HR** [9 students]

This round for me was only for around 20 mins since it was getting really late they had to finish it and take another round online but for starting 3 or 2 people they took it for almost 1 hr.

Unlike other rounds this wasnt that restrictive it was pretty much chill and the HR was also kind and friendly She started by asking me to introduce myself without any tech aspect and anything which is not there in my resume (So here i would say try speaking something which would be cool and might impress them so i spoke about my part in my schools kho kho team and about my bharatnatyam career)

Then asked Various questions like

1. Being a women if your idea didnt reach the higher management what would you do
2. Have you ever reached a meltingPoint, if yes then how did you deal with it
3. Why did you take CSE-AIML
4. A policy that you would want to be made compulsory by the company for all the women working
5. In 5 or 10 years where do you see yourself

There were few more women based situationship based question (try being as diplomatic as you can)

4. **Technical** [9 students]

So they decided to take another round online and this was again pretty much chill

Questions included are:

1. Introduce yourself
2. Why havent you done an internship
3. Was asked to explain one of my PowerBI project
4. Then asked where was each of the techs mentioned was used in the 3 of the projects that was there in my resume
5. Was all the mentioned fundamentals a part of your curriculum and if not which all
6. Timeline of my mentioned projects

Again it ended with me asking a question but since mine was AI related i had no idea if it would be appropriate to ask since it happened so fast like right after my HR it was suddenly said there would be an online interview and i would be the 1st one to be interviewed, i forgot to ask what kind of interview this was gonna be or was the interviewer HR or what , so here i first asked what exactly was his role and then after understanding he was also from a tech background i put my question forward

One thing i would advice is before asking your question first think properly if it is a proper question for your interviewer it would be kind of weird moment for you if he/she wont be able to answer.

8 students got accepted and this result was announced next day

Useful Tips:

So i know the placement season is a very hectic and tiring period but dont give up this was my third interview before this i didnt even pass few of the OAs and had also felt really disheartened when i got rejected in MSCI since it was one the company i had aimed for and dont end up taking a company that you had never wished for since i am sure you might end up regretting later, atleast in the start there might be few companies that is very

1. In all your interviewees keep on asking their name coz though it didnt happen for me but for few they had asked who had interviewed them before
2. Be at your best behaviour and without realizing we end up doing those faces, like whenever we dont know an answer or even when we are definitely sure of an answer coz they do judge on basis of everything so keep a control on that as well.
3. Be well versed with your intro and the question to be asked be ready with that try to match it with your interest or the companies tech related, i always asked some question that was AI related or some projects where they were including AI or something like that so be prepared for that as well.
4. Whatever i had learnt i initially didnt make notes be it for my DSA or SQL or anything keep it ready so as soon as a company comes you then give time to prepare for researching about the company or anything extra you might need for preparing and not wasting on learning something which you had already did before.
5. And for internship here i was able to give a reason but it might not be acceptable for many people i thought if its unpaid then i might not get to learn anything its not true, but for that sake dont even pay for those fake internship there you get to do only those projects which are available on the youtube you get no industrial level experience
6. Before your interview dont keep on mugging up things coz when you try to learn in the last moment you end up messing other things which you would be well versed with, so calm down listen to some music and just revise whatever you have learnt so far.
7. And mainly i repeat dont get sad because you couldnt clear the interview ofc there you should have strong fundamentals but it also depends on luck too the only thing in your hand is to clear the OA the rest of the things you cant predict what happens next.
8. Keep practicing don't let your one failure mess up your other possible victory.

Resources:

1. DSA:

For Python peeps i would say do neetcode sheet coz thats the main sheet i did striver is also good but stick to one and try completing as many concepts you can and majorly focusing on the important ones

2. SQL:

Complete the leetcode 50 this should be enough.

Extra: <https://www.interviewbit.com/courses/databases/sql-queries/>

 **SQL Window Function | How to write SQL Query using RANK, DENSE RANK, LEAD/LAG | SQL ...**
(For Advance SQL)

3. Puzzles:

▶ 50 Puzzles Commonly asked in HR Interviews (Covers most of the important puzzles)
<https://www.geeksforgeeks.org/aptitude/puzzles/> (More detailed and contains many more wont advice you to complete it all)

4. Guesstimates:

▶ Guesstimate Interview Question | Estimate Google Maps Storage | Technomanagers
(Not required for this role)

5. JAVA

▶ What is JDK, JVM, JRE, javac, Byte code, JIT ? Complete Execution Process !
<https://www.geeksforgeeks.org/java/output-of-java-program-set-4/>
(For OA) similar look at the rest of the sets and also of CPP based on the OA questions

6. OA

<https://whimsical.com/5-aptitude-Ar4hZFm8c8HnuReyBL9f4R>
(Contains all the links to indiabix and youtube videos necessary for clearing any logical, verbal reasoning kind of questions and it is more than enough)

7. OOPs

<https://whimsical.com/object-oriented-programming-cheatsheet-by-love-babbar-YbSgLatbWQ4R5paV7EggFw>
(contains of both CPP and JAVA)

And ofc chatgpt 😊

Feel free to reach out to me on my [Linkedin](#) or 9288000601

Last thing i would say is stay calm if you have put in your efforts then there is definitely a good company waiting for you.
All the Besttt!!